INCH-POUND

MIL-DTL-82968(OS)
23 September 1998

DETAIL SPECIFICATION

CUTTER, EXPLOSIVE ACTUATED, MK 33 MOD 0

This specification is approved for use by Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the MK 33 MOD 0 Explosive Actuated Cutter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standard, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

RR-W-410	Wire Rope and Strand

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1377

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STANDARDS

MILITARY

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-810	Test Method Standard for Environmental Engineering Considerations and Laboratory Tests
DOD-STD-2101	Classification of Characteristics

2.2.2 Other Government documents, drawings, and publications. The following other Governmental documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

DL 6784468	Cutter, Explosive Actuated, MK 33 Mod 0
6570660	Detonator, Non-Electric (Single)
6570661	Detonator, Non-Electric (Dual)
6702805	Initiator, Pyrotechnic Lead

(Copes of other Government documents, drawings, and publications required by contractors in connections with specific acquisition functions should be obtained from or as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742	Standard Practice for Radiographic Examination

(Application for copies should be addressed to the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated specifications or specification sheets), the text of the document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item definition. The MK 33 MOD 0 Explosive Actuated Cutter (cutter)(see 6.7.1) is approximately 9 inches long and 2.88 inches wide at its cutting base. The cutter weighs approximately 5 pounds in air and 4 pounds in water. The cutter consists of a body assembly, a safety block, and two cavities for Detonators. The external surfaces are black (see 3.3). The cutter is capable of being operated under water pressure of 28 psig to completely sever a one inch diameter plow steel wire rope.

3.2 First article inspection. When specified (6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Conformance to documents. The MK 33 Mod 0 Explosive Actuated Cutter covered by this specification shall be manufactured in accordance with DL 6784468 and all documents listed thereon.

3.4 Materials. All materials used in the manufacture of the cutters shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom, has been obtained from the cognizant Navy design activity prior to manufacture. When alternate material or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.5 Cutter inspection.

3.5.1 Pin assembly. Following assembly of the cutter blade and pin, the cutter blade edge shall rest securely within the notch of the pin (see 4.4.1).

3.5.2 Environmental requirements.

3.5.2.1 Transportation vibration. The cutter shall meet all acceptability requirements after being subjected to transportation vibration testing (see 4.4.4.1).

3.5.2.2 Salt fog. The cutter shall meet all acceptability requirements after being subjected to the salt fog test (see 4.4.4.2).

3.5.2.3 Rough handling drop-test. The cutter shall meet all acceptability requirements after being subjected to the rough handling drop-test (see 4.4.4.3).

3.5.3 Radiographic requirements. The cutters shall be free of imperfections when radiographed (see 4.4.2).

3.5.4 Ballistic requirements. The cutter shall meet ballistic requirements when subjected to the test prescribed by 4.5.1. The cutter shall be capable of being operated under water to completely sever a one inch diameter plow steel wire rope (see 4.5.2). The one inch diameter wire rope shall be Type I, Class 2, Construction 2 (6 x 19 Warrington) or Construction 3 (6 x 19 Seale), IWRC (Independent wire-rope core)

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6 X 7 wire rope with wire-strand core, improved plow steel, uncoated, preformed in accordance with Federal Specification RR-W-410.

3.6 Workmanship. The cutter shall be free of the following defects: poor or incorrect markings, burrs, dents, rust, corrosion, scratches to the black coating, damaged O-rings, chips, dirt, or other defects which could adversely affect ballistic performance. Any cleaning methods shall not be injurious to any of the components nor shall components be contaminated by cleaning agents employed. There shall be no chemical or electro-chemical corrosive or deteriorative effects from the manufacturing process.

4. VERIFICATION

4.1 Classification of inspections. The inspections herein are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2), first article inspection shall consist of the tests and examinations listed in table I. The first article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as will be used for regular production and shall be from the intended sources of supply. Any quantity production prior to notification by the contracting agency of the first article sample acceptability will be at the contractor's risk.

TABLE I. First article Tests.

Examinations and Tests	Sample											
	1	2	3	4	5	6	7	8	9	10	11	12
Visual Examination (3.3, 3.4, 3.6, 4.2, 4.4.3)	X	X	X	X	X	X	X	X	X	X	X	X
Radiographic Examination (4.4.2)	X	X	X	X	X	X	X	X	X	X	X	X
Pin Assembly (4.4.1)	X	X	X	X	X	X	X	X	X	X	X	X
Transportation Vibration Test (4.4.4.1)	X	X	X									
Salt Fog Test (4.4.4.2)				X	X	X	X					
Rough Handling Drop Test (4.4.4.3)								X	X	X		
Function Test (4.5.1)	X	X	X	X	X	X	X	X	X	X		
Retain Samples											X	X

4.2.1 Sample size. The first article sample size shall consist of twelve (12) assembled cutters, nine (9) sets of unassembled parts, and one (1) set of unassembled parts without coating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and

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procedures as used in regular production. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawing and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.5).

4.2.2 Inspection routine. Ten (10) cutters shall be expended in the tests listed in Table I and two (2) shall be retained for investigative purposes.

4.2.3 Failure. Failure of the sample to meet the requirements of 3.3 through 3.6 shall be cause for rejection of the first article sample.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of verification of the cutter characteristics and requirements of DL 6784468 and the tests and examinations of Table II.

TABLE II. Quality Conformance Tests and Examinations.

Item	Nature of Test	Requirement	Test	Classification (DOD-STD-2101)
Cutter	Visual Examination	3.6	4.4.3	(M101)
Cutter	Radiographic Examination	3.5.3	4.4.2	
Cutter	Pin Assembly	3.5.1	4.4.1	(M102)
Cutter	Ballistic Requirements	3.5.4	4.5.1	(M103)
Cutter	Preservation-packaging, Packing and Marking	5.1	4.4.3	(M104)

4.3.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 2 and not greater than 3000. Cutters required for all test purposes shall be selected from the production lot and shall not be applied as part of the quantity specified for delivery by the contract.

4.3.2 Sampling. A sample of cutters for ballistic test shall be selected from each production lot in accordance with table III. In addition to the sample of cutters for ballistic test, two (2) additional cutters shall be furnished from each production lot to be retained for investigative purposes.

4.3.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M).

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The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

TABLE III. Quality Conformance Sample Size.

Lot Size	Sample Size	Retained Sample
20 - 500	10	2
501 - 1000	15	2
1001 - 1500	20	2
1501 - 2000	25	2
2001 - 2500	30	2
2501 - 3000	35	2

4.4 Test.

4.4.1 Pin assembly inspection. One hundred percent examination after cutter blade and pin assembly shall demonstrate that the cutting edge of the cutter blade is resting in a seated position within the notch in the pin. An assembly wherein the cutting edge of the cutter blade is not in seated position within the pin notch shall be considered defective. Cutters that fail to meet the requirements of 3.5.1 are defined as defective.

4.4.2 Radiographic examination. All first article and lot acceptance test samples shall be radiographically examined in accordance with ASTM-E1742. Any observable imperfections in components or assembly shall be cause for rejection. If any assembly or component defects are noted, the contractor shall 100 percent X-ray the represented lot and submit the radiographs to the design activity for review. The cutters shall be positioned for the most revealing view with the long axis perpendicular to the x-ray beam. The radiographs shall be in consecutive order and each radiograph shall carry a permanent identification of the cutters displayed thereon. The radiograph identification shall include item nomenclature, the complete lot number, contract number and complete span of test control numbers displayed thereon. Any missing test control numbers shall be identified on the radiograph. If radiographs of the entire production lot are required they shall be at the expense of the contractor. Defective assemblies found during radiographic review shall be removed from the production lot.

4.4.3 Visual examination. All cutters shall be visually examined. Cutters having any of the defects of 3.3, 3.4, 3.5.1, and 3.6 shall be rejected. The preservation-packaging, packing, and marking of the cutter shall be examined for conformance to section 5.

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4.4.4 Environmental tests.

4.4.4.1 Transportation vibration test. The cutters, enclosed within their shipping containers, shall be subjected to transportation vibration test B2 of MIL-STD-331 at 23 ± 10 °C. Defectives are cutters that fail to meet the requirements of 3.5.2.1.

4.4.4.2 Salt fog test. The cutters, outside of their shipping containers, shall be subjected to a forty-eight hour salt fog test defined in MIL-STD-810, Test Method 509.3. After completion of the test, the cutters shall meet the requirements of 3.3 through 3.5. Defectives are cutters that fail to meet the requirements of 3.5.2.2.

4.4.4.3 Rough handling drop test. The unpacked cutter shall be dropped, in random orientations, onto a non-yielding surface. Four drops from 36 inches and eight (8) drops from 18 inches. Defectives are cutters that fail to meet the requirements of 3.5.2.3

4.5 Ballistic tests.

4.5.1 Function test. The cutter shall be functionally tested in a suitable open tank capable of holding at least three feet of water. The tank shall include internal means for supporting the cutter. Prior to testing, a nine inch length of one inch diameter plow-steel wire rope per RR-W-410 shall be secured by the cutter spring below the cutter blade. One (1) Drawing 6570661 Non-Electric (Dual) Detonator or two (2) Drawing 6570660 Non-Electric (Single) Detonator shall be installed in the firing head and secured by the cap. The cutter shall then be mounted within the tank no less than two feet beneath the water surface and then the detonators shall be fired by means of a Drawing 6702805 Pyrotechnic Lead Initiator. Cutters that fail to meet the requirements of 3.5.4 are defined as defective.

4.5.2 Acceptance criteria. All cutters subjected to the ballistic tests of 4.5.1 shall meet the requirements of 3.5.4. A failure of any cutter to meet this requirement shall be cause for rejection of the lot represented.

4.6 Retest. There shall be no retests.

4.6.1 Test failure. If test failure is attributable to an assignable cause, excluding the cutter, original test results should be discarded and that portion of the test reconducted.

5. PACKAGING

5.1. Packing. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cutter is an underwater component used by a diver for ballistically severing a one inch diameter plow steel wire rope (cable) under seawater at a pressure of 28 psig. This cutter is used for special military underwater operations (i.e., by the SEALS). There is no commercial application for this design.

6.2 Acquisition requirements. Acquisition document must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Item definition (see 3.1 and 6.7.1).

d. Whether first article sample is required, and if so, specify the test activity (see 3.2).

e. First article inspection continuity (see 4.2 and 6.6).

f. Sample size (see 4.2.1 and 6.5).

g. Production lot size and test activity (see 4.3.1).

h. Items of data required for each first article and production lot.

i. Directions for shipping ballistic sample to the activity conducting production lot acceptance test.

j. That the safety precaution requirements of the “Contractor’s Safety Manual for Ammunition, Explosives, and Related Dangerous Material, “DOD 4145.26M are applicable. NOTE: When this Specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of “Ammunition and Explosives Ashore,” OP5 are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable referenced documents in section 2 or (c) referenced lower-tier documents, need not be prepared for the government unless specified in the contract of order. The data to be furnished should be listed on DD Form 1423 (Contract Data Requirements List) which should be attached to and made part of the contract or order. NAVSUP Form 4200/15 (Drawings, Lists, and Specifications Required) should be attached where applicable.

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6.4 Ammunition Data Card. Data cards should be prepared in accordance with MIL-STD-1168 and should state the quantity and destination.

6.5 First Article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of twelve (12) assembled cutters, nine (9) sets of unassembled parts, and one set of unassembled parts without coating or final protective finish. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 First Article Inspection Continuity. The contracting officer should insure that the first article inspection be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used for regular production and should be from the intended sources of supply. Any quantity production prior to notification by the contracting agency of the first article sample acceptability will be at the contractor's risk.

6.7 Definitions.

6.7.1 Cutter. The term "cutter" means the Cutter, Explosive Actuated, MK 33 MOD 0, DL 6784468 (see 3.1 and 6.2).

6.8 Conflicting Requirements. Refer in writing to the procuring agency for clarification, any conflicting requirements arising between this specification and any specification, publication, or drawing listed herein.

6.9 Subject term (key word) listing.

Ballistic test
Cable cutter
Detonator, non-electric
Diver equipment
Pyrotechnic initiator
Submerged component
Wire rope

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Custodians:
Navy - OS

Preparing activity:
Navy - OS
(Project 1377-0095)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-82968(OS)	2. DOCUMENT DATE (YYMMDD) 98-09-23

3. DOCUMENT TITLE
CUTTER, EXPLOSIVE ACTUATED, MK 33 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office, Attn: DLSC-LM 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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